

WAHPETON, ND, USA

WMO: 722004

Lat: 46.244N

Lon: 96.607W

Elev: 968

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-14.7	-25.8	1.4	-19.0	-19.9	1.9	-13.7	31.4	13.5	28.0	13.0	5.9	320	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	88.5	71.8	85.8	69.9	82.2	68.2	76.3	84.0	74.0	82.1	71.7	79.3	13.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.6	80.5	71.9	122.2	79.2	69.6	112.8	77.4	40.5	84.2	38.2	82.2	36.1	78.4	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.0	24.4	21.0	-26.1	94.2	6.4	3.1	-30.7	96.5	-34.4	98.3	-38.0	100.0	-42.6	102.3		
			-27.0	79.5	6.4	3.7	-31.6	82.2	-35.3	84.4	-38.9	86.4	-43.5	89.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	10.8	10.9	28.0	42.6	56.5	66.0	70.5	67.5	60.1	45.6	31.2	15.5
	DBStd	23.79	14.25	13.12	13.26	10.48	9.26	6.06	5.54	5.64	8.27	9.03	11.22	12.54
	HDD50	5144	1214	1096	689	259	42	0	0	0	13	196	567	1068
	HDD65	8766	1679	1516	1148	673	294	60	16	38	190	606	1013	1533
	CDD50	2322	0	0	6	36	243	480	634	542	317	60	4	0
	CDD65	467	0	0	0	0	30	90	185	114	44	4	0	0
	CDH74	4581	0	0	1	12	402	836	1777	1022	469	62	0	0
	CDH80	1286	0	0	0	2	132	203	535	273	125	16	0	0
(14) Wind	WSAvg	9.7	10.2	10.1	10.6	11.2	10.9	9.2	7.5	7.5	9.2	10.4	10.2	9.8
(15) Precipitation	PrecAvg	22.1	0.6	0.4	0.9	2.1	2.9	3.8	3.3	2.7	2.4	1.8	0.7	0.5
	PrecMax	36.3	2.4	1.2	2.7	7.6	7.9	8.8	8.3	8.5	7.3	5.7	1.8	2.0
	PrecMin	8.3	0.0	0.0	0.2	0.1	0.4	0.7	0.4	0.3	0.3	0.1	0.0	0.0
	PrecStd	5.0	0.6	0.3	0.5	1.5	1.9	2.1	1.9	1.6	1.7	1.5	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	46.3	66.3	74.8	90.0	90.2	92.8	90.6	89.8	81.7	63.7	45.0
		MCWB	35.9	42.5	55.3	55.8	64.7	69.2	77.9	72.5	71.8	63.6	52.2	39.5
	2%	DB	35.7	36.5	57.4	69.8	83.7	85.7	88.7	86.4	83.5	72.4	57.1	36.7
		MCWB	33.8	33.8	50.5	53.5	63.4	67.6	75.2	72.3	67.5	57.5	48.4	33.6
	5%	DB	32.5	33.7	51.5	63.9	78.7	81.6	85.8	82.2	80.5	64.4	51.8	34.0
		MCWB	31.6	32.2	46.3	51.2	59.7	65.2	72.3	70.0	65.7	54.1	45.6	32.6
	10%	DB	30.4	30.5	44.9	59.4	72.8	79.1	82.1	79.8	75.0	60.7	45.7	32.0
		MCWB	29.8	29.5	41.2	48.1	57.9	64.7	69.9	67.4	63.7	51.0	41.2	31.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	44.4	57.4	60.3	72.7	76.3	82.4	77.9	75.8	67.3	55.1	40.2
		MCDB	39.1	47.6	62.6	70.4	81.3	84.2	90.4	83.9	85.4	77.3	61.5	44.3
	2%	WB	33.4	34.3	50.9	55.8	66.2	72.5	77.8	74.7	71.6	59.5	49.5	34.3
		MCDB	34.4	36.2	56.5	65.5	75.7	79.7	84.9	82.4	78.9	71.0	56.6	36.2
	5%	WB	31.8	31.8	45.3	52.8	63.3	69.5	75.0	72.5	67.7	55.4	45.4	32.4
		MCDB	32.5	32.8	49.5	61.6	71.4	76.2	82.6	80.4	75.3	62.6	51.4	33.3
	10%	WB	29.9	29.4	40.7	49.9	60.4	67.0	72.4	70.1	65.0	52.1	41.3	30.8
		MCDB	30.6	30.6	45.1	58.4	70.8	75.6	80.4	77.0	73.1	58.9	45.7	31.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.9	15.8	18.9	21.6	20.0	21.4	21.0	22.0	18.5	16.1	15.3
		MCDBR	17.0	18.9	23.8	28.6	29.7	24.5	24.0	24.1	27.6	27.3	23.9	17.4
		MCWBR	15.7	16.8	16.2	15.7	14.4	11.8	12.4	12.3	13.4	15.7	15.9	14.9
	5% WB	MCDBR	16.2	18.2	22.1	25.1	22.8	20.0	22.4	21.5	22.8	23.7	22.9	15.7
		MCWBR	15.0	16.7	15.6	15.5	13.4	12.1	13.4	12.7	13.9	14.7	15.6	13.8
(40) Clear-Sky Solar Irradiance	taub		0.297	0.296	0.309	0.361	0.381	0.390	0.389	0.391	0.359	0.318	0.286	0.267
	taud		2.264	2.323	2.423	2.331	2.346	2.367	2.357	2.346	2.424	2.514	2.510	2.415
	Ebn at Noon		250	279	292	284	281	278	277	271	272	269	260	254
	Edh at Noon		25	30	31	37	38	38	38	36	31	24	20	20
(44) All-Sky Solar Radiation	RadAvg		540	895	1220	1554	1749	1955	2056	1723	1308	820	519	388
	RadStd		56	96	117	120	120	121	81	153	84	113	58	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page